

Manual

SMA-PRJ Project Time Recording SMA-PRJ 8.1

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Contents

1	SMA-PRJ Project Time Recording.....	4
2	Project Time Recording.....	6

1 SMA-PRJ Project Time Recording

Purpose

You can use this function package to record the hours worked on project orders via SMA.

Implementation notes

You use the function package for the following purposes:

- Employees can record the hours worked on project orders.
- The working times are not recorded on the terminal via login and logout; instead an absolute time is entered as working time.
- You want to record the hours via SMA.

Integration

This function package is based on the HYDRA shop floor data collection (function package *Shop Floor/Order Data Management*).

The released project orders can be passed from an ERP system via the HYDRA ERP interface** or from SAP PS via the SAP interface CA-PDC / CC4**. Another option is to manually create project orders on the MES Operation Center (MOC).

You can upload the times recorded to an ERP system via the HYDRA ERP interface**. You can upload the times to SAP PS via the SAP interface CA-PDC / CC4**.

** Requires additional licensing.

Features

- Display of the pool of orders including all project orders assigned to an employee for which you can record times.
- Input dialog to record times for specified projects (orders). Booking of the recorded data in HYDRA.
- Evaluation and display of the recorded data (times) including cancellation function (the log records generated for the data recorded can be deleted).
- You can evaluate or edit the recorded data on the MES Operation Center (e.g. order information, order overview, editing of postings), if the license for the function package *Shop Floor/Order Data Management* is available.

2 Project Time Recording

App name	Project time recording
Function authorization	sma.prj

Purpose

Use this application to upload times for project orders. Uploads document the status of processing and provide a forecast on how the project will develop. You must upload exact data to ensure realistic and exact project planning and project tracking.

You may enter the following data for uploads:

- Date/Time
- Actual duration
- Indicator of the upload status (partial upload, final upload)

Integration

The recorded uploads are integrated in BDE (shop floor data collection) and can be evaluated in most BDE functions of the MOC. In addition, the recorded postings can be uploaded to a higher-level system.

Configuration

You must make the following configurations, if you want to record project times:

Workplace configuration

If you plan to upload data for an operation, you must configure the workplace where this operation is planned as follows:

Enable the option *Posting on OPs that are not logged on*.

Sequencing list: The sequencing list should be set to "M" (pool of machines/workplaces) or "H" (group control). Also take into account the notes on the planning process in the section *Work plan*.

Assign a responsibility area to the workplace.



For these workplaces, ONLY project times may be uploaded via SMA. Postings like "log OP on" or "log OP off" in the terminal or MOC are not supported.

HR master

Create a person in the HR master data.

Assign the person in the tab *Shop floor data* to the ("master") workplace for which the operations are planned. The person can then perform uploads for this workplace via the SMA application.

User administration

Create a user in the *User administration*.

Assign the person from the HR master data to this user.

Assign the responsibility area of the workplace to this user. The data of this workplace, where the operation is planned on, is uploaded via SMA Project time recording.

Assign the function authorization sma.prj to the user.

Work plan

Create the orders with a special order type (e.g. order type "5 - project order").

If you want to upload detailed data, then the order and the included operations should correspond to the individual activities that are carried out in the project or order.

You **must** plan each operation for a corresponding workplace, if you want to upload data for this operation. Therefore, the operations should either be planned (in detail) by the ERP system or by one of the planning functions provided by HYDRA.



If you perform planning using the HYDRA Shop floor scheduling (HLS) or Graphic order sequencing (GAV), you must specifically define the formula for the remaining run time: Here, the remaining run time must be calculated using the actual duration (time posted to RPA 11) and the result is therefore reduced.

Upload procedure using SMA

Proceed as follows to perform uploads.

1. Open the menu item "Pool of orders".
2. [Select operation](#) from the Pool of orders for which you want to perform uploads.
3. [Input of upload data](#).

These steps are described in more detail in the paragraphs that follow.



Using the menu item "Direct posting", you can enter data directly without selecting an operation from the pool of orders. In this case, specific input fields are not populated and must be filled manually. We recommend to enter data using the menu item "Pool of orders".

Select operation

The system identifies the workplace assigned to the person, selects all planned operations for this workplace and shows the operations in a list.



If no workplace is defined for the person in the HR master data, no data will be shown.

The following data is shown in ascending order by MES order number.

Field	Meaning
MES order number	Combined order/OP number. The combined order/sequence/OP number is shown here if the system is configured for sequence processing.  Note that in general only orders/operations of the order type 5 are displayed irrespective of the entered selection criteria.
Project	Project number
Article/Item	Article number (as defined in the operation)
Article name	Article name
OP designation	Operation name/designation
Workplace	Workplace for which the operation is planned.
Group	Group of the workplace for which the operation is planned.

Select the operation for the upload and click the button "Enter posting".

Reduce list of pool of orders

The filter function  offers the possibility to reduce this list by selecting the fields described below. All fields support wildcard characters. Please consider case sensitivity.

- Search by the MES order number
- Search by the project number
- Search by the operation's article

- Search by the operation's OP designation

Input of upload data

Enter your project times in the dialog "Enter upload".

If you have started the dialog via the list *Pool of orders*, the fields *MES order number* and *Workplace* are automatically populated using the data from the pool of orders. In addition, the date and time fields are filled with the current values.

Now enter the data for the upload:

Field	Comment
MES order number	Combined order/OP number. Enter the combined order/sequence/OP number here if the system is configured for sequence processing.
Workplace	If no workplace is entered, data is uploaded for the first workplace of the group for which the operation is planned.
Date	The current date is pre-allocated.
Time	The current time is pre-allocated.
Duration	Duration to be uploaded. You may use the following formats for the entry: • Hour:Minute, e.g. 2:42 • Hour,decimal minute, e.g. 2,7 • Hour.decimal minute, e.g. 2.7
Finish operation	Specifies whether or not the operation is finished. Set to "No" by default
Person	By default, the personnel number used for login is pre-allocated and cannot be changed.

Once the input has been confirmed by clicking the relevant button, the entered data are posted in the system and the dialog is closed.

Posting

The entered data are posted once confirmed in the dialog. It is an interruption posting as far as the system is concerned (DLG=A_UN|ANR=<MES Order Number>|MNR=<Workplace>|PNR=<Person-ID>|DAT=<Date>|ZEI=<Time>|EGR:BMK11=<Duration>|). Consequently, a BDE log record of record type U is generated in the system. However, the interruption posting will only be carried out if a duration greater than 0 is entered.

If the workplace has been configured as a "group workplace " (type = G), the system generates a personal BDE log record of record type B in addition to the order-related posting.

Der Service sendet den Parameter **OPT:AGISTPNR=G** in den Dialogdaten an den HYDRA-Leitrechner. Dadurch wird bei einer AG-IST-Meldung an GAP ein "B"-Satz für die meldende Person angelegt. Der "B"-Satz ist eine 1:1 Kopie des "U" oder "E"-Satz.



In general, only the order-related RPAs are posted. Labor utilization and staff-related resource performance accounts are not posted.

If the option "Finish operation" has been set to "Yes", a logoff posting (DLG=A_BE) is sent, once the duration has been uploaded for the operation (DLG=A_UN). The "Finish OP" posting results in the generation of a BDE log record of record type E and finishes the operation (status "finished"). The "Finish OP" posting is also carried out, if the entry for the duration is 0 and no upload was performed.

Once the operation has been logged off, nobody can upload data for the operation anymore.

In case of an interruption posting, you can enter or change the point in time (date, time) manually (DLG=A_UN). But in case of a finish posting (DLG=A_BE), the system assigns and posts the current point in time (of the server). The times of interruption posting and finish posting are therefore always different.

Posting to RPA

The recorded actual duration is posted to RPA 11 of the operation.

Start date/finish date

Start date and finish date are only used for informational purposes. These dates do not affect the actual duration. There is no plausibility check of the actual duration against the period between start date and finish date.

Start and finish date remain unchanged (there is no calculation of the finish date based on start date and duration).

Shift date/shift

Shift date and shift number, which are stored in the posting, are based on the entered start date. You must assign a valid shift model to the workplace in order to post data and thus to get the shift date and the shift number from the entered start date.



The data collected via the project time recording cannot be recalculated in the event maintenance. Consequently, you must maintain postings in the maintenance of postings.

Display and cancel uploads

You can view your recorded uploads using the menu item "Posting list". Enter the period of time for which you want to view the uploads (date from/to, by default <today> minus 7 days until <today>). The system identifies all data records from the start day as of 0:00 midnight until the end day 23:59.



Whether or not all uploads/postings can be shown for the specified period of time depends on their retention period configured in HYDRA. By default, uploads of the last 35 days are available.

After having requested the data, the uploads are shown that have been recorded for the workplace, which is part of the responsibility area you are authorized for.

The following data are shown in descending order by posting time (column *Start*).

Field	Meaning
MES order number	Combined order/OP number. The combined order/sequence/OP number will be shown here if the system is configured for sequence processing.
Article/Item	Article number (as defined in the operation)
Article name	Article name
OP designation	Operation name/designation
Workplace	Workplace for which the upload has been made
Duration	The duration posted onto RPA 11 from the recorded upload
Start	Posting time of upload
Record type	"Interruption of order" if the option "Finish operation" is set to "No" for the upload. "Finish order posting" if the option "Finish operation" is set to "Yes" for the upload.



The list of order-related postings includes the interruption postings (DLG=A_UN) as well as the finish postings (DLG=A_BE). The duration 00:00 is the easiest way to distinguish between finish postings and interruption postings.

You can cancel or delete an upload, if you select the respective upload in the list and click the button .

The action "delete" deletes the selected posting. If there is a corresponding posting of record type B, it will also be deleted.

If the "Finish" posting of an operation is deleted, the operation status is reset to "U = interrupted".



A posting can only be deleted, if it has not yet been uploaded to the ERP system.

The option "Change after upload" for the order type is not evaluated. That means, if a posting has been uploaded, it cannot be deleted, even if the option is set differently to the order type. In this case, you must delete the posting in the MOC application *Order-related postings*.

Evaluations in MOC

The times recorded and posted to the operation are shown in the following MOC applications. You need further authorizations to start the applications.

Order information

Shows the actual durations posted to RPA 11 of an operation.

Order overview → Progress

Shows the posted actual durations of the column RPA 11, just as it is the case for the "Status" tab of the order information dialog.

Order-related postings

The actual durations posted onto the resource performance account 11 can be displayed and changed in the maintenance of postings dialog.

Personnel report

If the workplace, for which you uploaded data, has been configured as group workplace, this report shows the actual durations posted to the RPAs.

Upload to the ERP system

The recorded data are uploaded to the ERP system. For further information on the upload structure, please refer to the respective interface documentation.

